

EXAM 7 – SPRING 2012

15. (2.75 points)

Given the following auto liability claims information:

<u>Claim #</u>	<u>Case Incurred Indemnity (\$)</u>	<u>Case Incurred ALAE (\$)</u>
1	800	100
2	5,000	250
3	2,100	0
4	12,000	1,800
5	77,000	4,000

a. (1 point)

Calculate the Pearson's correlation coefficient.

b. (1.25 points)

Using a dependency measure other than Pearson's correlation coefficient, draw a conclusion with respect to the appropriateness of using Pearson's correlation coefficient for this set of claims information.

c. (0.5 point)

Identify one similarity and one difference between Pearson's correlation coefficient and the dependency measure in b. above.

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Question 15 Sample Answer

Solution 1

- a) X – Case Incurred Indemnity
Y – ALAE

$$E(X) = 1/5(800+5000+2100+12000+77000) = 19380$$

$$E(X^2) = 1/5(800^2+5000^2+2100^2+12000^2+77000^2) = 1,220,610,000$$

$$E(Y) = 1/5(100+250+0+1800+4000) = 1230$$

$$E(Y^2) = 3,802,500$$

$$E(XY) = 1/5[(800(100) + 5000(250) + 2100(0) + 12000(1800) + 77000(4000)] \\ = 66,186,000$$

$$\sigma_X = [E(X^2) - E(X)^2]^{1/2} = 29069.32404$$

$$\sigma_Y = [E(Y^2) - E(Y)^2]^{1/2} = 1532.840501$$

$$\text{Pearson's } \rho = \frac{E(XY) - E(X)E(Y)}{\sigma_X \sigma_Y} \\ = \frac{66186000 - (19380)(1230)}{(29069.32404)(1532.840501)} \\ = 0.950402$$

- b) Use Kendall's Tau (τ)
- Rank the Indemnity (I)

<u>Claim #</u>	<u>Indemnity (I)</u>	<u>ALAE</u>
1	800	100
2	2100	0
3	5000	250
4	12000	1800
5	77000	4000

of swaps Q = 1 (swap 0 & 100)

$$\begin{aligned}\tau &= 1 - \frac{Q}{\# \text{ of pairs} / 2} \\ &= 1 - \frac{1}{10 / 2} \\ &= 0.8\end{aligned}$$

$$\begin{aligned}\# \text{ of pairs} &= n(n-1) / 2 \\ &= 5(4) / 2 \\ &= 10\end{aligned}$$

It can be concluded that the case incurred indemnity is positively correlated with the ALAE due to the high correlation measure from Pearson's ρ and Kendall's τ (both close to 1)

- c) Similarity: Both summarize the overall dependency in one single number.
Difference: Pearson's correlation uses value in determining correlation whereas Kendall's τ uses ranks in determining correlation.

Solution 2

- a) Pearson's $\rho = \text{Cor}(x,y) / (\sigma_X \sigma_Y) = .95$

$$\text{Cor}(x,y) = E(XY) - E(Y)E(X)$$

$$\sigma^2_X = E(X^2) - E(X)^2$$

$$\sigma^2_Y = E(Y^2) - E(Y)^2$$

$$E(X) = (800+5000+2100+12000+77000)/5 = 19380$$

$$E(X^2) = (800^2+5000^2+2100^2+12000^2+77000^2)/5$$

$$E(Y) = 1230$$

$$E(Y^2) = (100^2 + 250^2 + 0^2 + 1800^2 + 4000^2) / 5$$

$$E(XY) = [(800(100) + 5000(250) + 2100(0) + 12000(1800) + 77000(4000)]/5$$

- b) Spearman's $\rho = 1 - (6s/(n(n^2-1)))$ where s = sum of squares of difference in ranks

<u>Claim</u>	<u>Indem</u>	<u>ALAE</u>	<u>Indem Rank</u>	<u>ALAE Rank</u>	<u>(I-rank - A-rank)²</u>
1	800	100	1	2	$(2-1)^2 = 1$
2	2100	0	3	3	$(3-3)^2 = 0$
3	5000	250	2	1	1
4	12000	1800	4	4	0
5	77000	4000	5	5	0
					<u>2</u>

$$\text{Spearman's } \rho = 1 - \frac{6(2)}{5(25-1)} = .9$$

Pearson's appears to appropriate since it is similar to Spearman's. One of the Pearson's weaknesses is that it is severely affected by outliers but the entire distribution appears correlated so it's not a problem here.

- c) Similarity: both weight by the squares of the differences
 Different: Spearman is an ordinal measure of correlation where as Pearson is a cardinal measure

Solution 3

a)

$$\text{Pearson's } \rho = \frac{\sum \tilde{x}_i \tilde{y}_i}{\sqrt{\sum \tilde{x}_i^2 \sum \tilde{y}_i^2}} \quad \begin{array}{l} \tilde{x}_i = x_i - \bar{x} \\ \tilde{y}_i = y_i - \bar{y} \end{array}$$

Claim	Incurred (x_i)	ALAE(y_i)	$\tilde{x}_i$	$\tilde{y}_i$
1	800	100	-18580	-1130
2	2100	0	-14380	-980
3	5000	250	-17280	-1230
4	12000	1800	-7380	570
5	77000	4000	57620	2770

$$\bar{x} = 19,380 \quad \bar{y} = 1230$$

$$\sum \tilde{x}_i \tilde{y}_i = 2.117 \times 10^8$$

$$\sum \tilde{x}_i^2 = 4.225 \times 10^9 \quad \sum \tilde{y}_i^2 = 11,748,000$$

$$\therefore \rho = 0.95$$

b) I will use Spearman's ρ

$$\text{Spearman's } \rho = 1 - \frac{s}{\left(\frac{n(n^2-1)}{6} \right)} = 1 - \frac{2}{\left(\frac{5(25-1)}{6} \right)} = 0.9$$

$$s = \sum (\text{difference of ranks})^2 = 1^2 + 0^2 + (-1)^2 + 0^2 + 0^2 = 2$$

Claim	rank of incurred	rank of ALAE	Difference of rank
1	5	4	1
2	3	3	0
3	4	5	-1
4	2	2	0
5	1	1	0

Conclusion: Pearson is heavily affected by outliers. In this case the effect was not so bad because the huge case incurred was in the “right” direction i.e. larger ALAE as well as larger Incurred. If it had been in the opposite direction, i.e. much smaller incurred; it would have dragged the correlation down and possibly made it negative.

I would not recommend Pearson’s for this data is much better to use either Spearman’s ρ or Kendall’s τ which focus on order not values.

- c) Similarity: They both summarize the correlation in a single value.
 Difference: Pearson’s correlation focuses on the values of the data and their distance from the mean. Spearman’s correlation focuses on the order of the data points and their ranks.

Examiner Comment

Candidate can calculate either Kendall’s Tau (0.80) or Spearman’s (0.90) rank correlation. Both give results similar to Pearson, that is, strong positive correlation. Candidates should conclude that Pearson’s correlation is appropriate in this case. We also give credit if they say that Pearson’s may be appropriate, but the 77K outlier is driving up the correlation since it’s not truly linear. Basically any reasonable conclusion based on assumptions is valid for full credit. If candidate’s comment was a correct conclusion based on incorrect calculation, we will give full credit for the comment.